

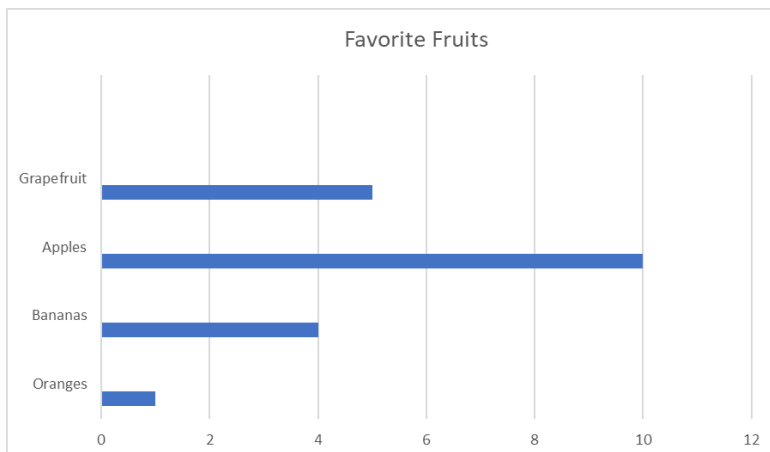


Mathematics Written Assessment 2023 Elementary District Governor's Cup

Student Instructions:

- For each question, choose the BEST answer.
- What is the value of 26 more than 10^2 ?
 - 2600
 - 1026
 - 1260
 - 126
 - Which set of numbers is not arranged from greatest to least?
 - 1789, 1790, 1792, 1795
 - 154019, 153019, 152019, 151019
 - 1790, 1789, 1788, 1787
 - 15, 11, 0, -4
 - What is the value of the underlined digit: 147.439
 - 3 tenths
 - 3 hundredths
 - 3 thousandths
 - 3 ten-thousandths
 - Evans Chebet won the Boston Marathon in 2022 with a time of one hundred twenty-six and eighty-five hundredths' minutes. What number below shows his time correctly?
 - 126.85
 - 12.685
 - 1.2685
 - 1268.5

Use the Bar Graph to solve questions 5, 6, and 7. The numbers on the X-axis represent number of people.



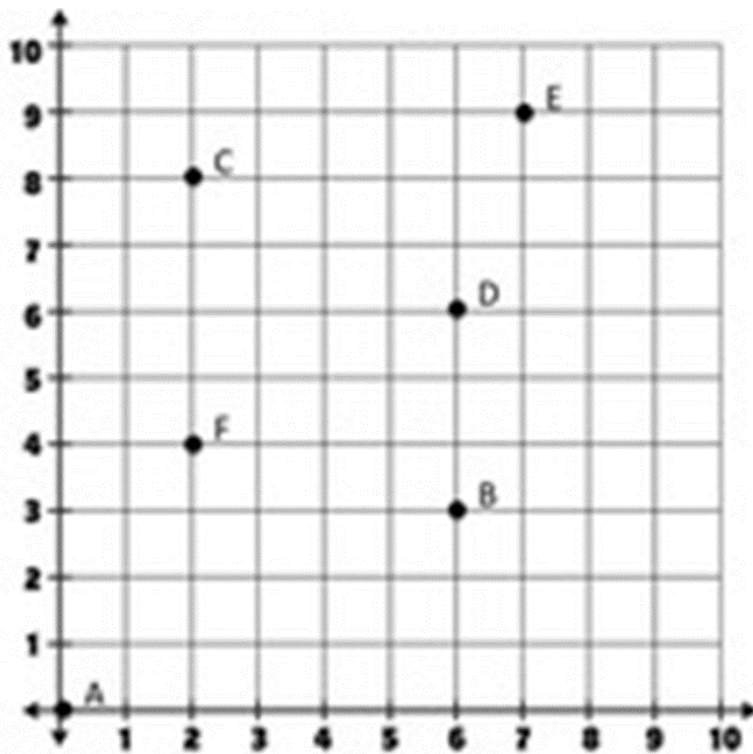
5. How many total people chose apples and grapefruit as their favorite fruits?
- A. 10
 - B. 4
 - C. 15
 - D. Not able to be determined from information given
6. Based on the graph, which fruit is the least favorite?
- A. Grapefruit
 - B. Apples
 - C. Bananas
 - D. Oranges
7. How many more people prefer apples to bananas?
- A. 10
 - B. 4
 - C. 6
 - D. 14
8. You want 10 tickets to ride at the carnival. The tickets cost \$1.25 each if you buy them separately or you can buy a book of 5 tickets for \$5.00. How much money will you save by buying the 2 books of tickets?
- A. \$12.50
 - B. \$10.00
 - C. \$5.00
 - D. \$2.50
9. What is the value of $\frac{3}{8}$ converted to a decimal?
- A. 0.375
 - B. 03.75
 - C. 375.0
 - D. 2.6666
10. Karen spent $\frac{1}{5}$ of an hour working on math homework, $\frac{3}{10}$ of an hour working on Spanish, and $\frac{1}{2}$ hour working on her art project. How much time did she spend working on her homework?
- A. $\frac{3}{100}$ of an hour
 - B. 1 hour
 - C. $\frac{5}{17}$ of an hour
 - D. 2 hours
11. What are all the positive factors of 36?
- A. 1, 2, 3, 4, 9, 12, 18, 36
 - B. 6 and 6
 - C. 1, 2, 3, 4, 6, 9, 12, 18, 36
 - D. 36 and 1

12. The school is 1.5 blocks from Kelly's house. The library is 1.25 blocks from the school and the gym is 3.1 blocks from the library. The gym is 2.4 blocks from Kelly's house. Kelly walks to the gym and then to the library and decides to go by the school on his way home. How many blocks will he walk to get back home?
- A. 19.5 blocks
B. 8.25 blocks
C. 1.95 blocks
D. 82.5 blocks
13. Jennifer decided to give away her Pokémon cards. She gave 8 cards each to 5 friends and she had 17 cards left. How many cards did she have before she divided them?
- A. 30
B. 23
C. 77
D. 57
14. The restaurant pays \$10.50 per hour for the first 6 hours and \$12.50 per hour for any hours after that in a shift. Lin worked 6 hours on Friday, 7 hours on Saturday, and 6 hours on Sunday. How much money did he earn for the three days?
- A. \$63.00
B. \$37.50
C. \$201.50
D. \$213.50
15. The two kindergarten classes are making valentines for two nursing homes. One class makes 42 valentines and the other makes 35. How many valentines will be left over if they divide the valentines equally between the nursing homes?
- A. 0
B. 1
C. 38.5
D. 77
16. The academic coach kept track of the number of questions answered correctly during Quick Recall. He used the stem and leaf plot to record the data. On how many days did the team answer more than 30 questions but less than 60 questions correctly?

Questions Answered Correctly	
Stem	Leaf
2	9 9
3	
4	9 8 7
5	1 1 1
6	5 1 3

- A. 6 days
B. 5 days
C. 4 days
D. 3 days

17. Triangle ABC is congruent to triangle DEF. Which angle is congruent to angle A?
- A. None are congruent
 - B. Angle D
 - C. Angle E
 - D. Angle F
18. The bakery is selling birthday cakes for $\frac{1}{3}$ off the regular price. If a chocolate cake costs \$24.00 at the regular price, what will be the sale price?
- A. \$ 24.33
 - B. \$ 16.00
 - C. \$ 16.33
 - D. \$ 8.00
19. A runner reached a speed of 44.72 kph in 2009. The crew of a rocket reached a speed of 39,897.51 kph in 1969. How much faster did the rocket's crew travel?
- A. -39,852.79 kph
 - B. -34,026 kph
 - C. 34,026 kph
 - D. 39,852.79 kph
20. Match the objects with an appropriate measurement.
- | | |
|------------------|----------------------|
| x. Orange juice | 1. Kilometer |
| y. Highway | 2. Meter |
| z. Doghouse roof | 3. Cubic centimeters |
- A. $x - 2, y - 1, z - 3$
 - B. $x - 1, y - 2, z - 3$
 - C. $x - 2, y - 3, z - 1$
 - D. $x - 3, y - 1, z - 2$



Use the coordinate plane shown to answer questions 21, 22, and 23.

21. Which point on the plane is located at (7, 9)?
A. Point C
B. Point D
C. Point E
D. Point F
22. What are the coordinates of point A?
A. Not given
B. (0,0)
C. (1,1)
D. (0,1)
23. Which of the following does not describe ways to move from point F to point B?
A. Down 1, Right 4
B. Down 4, Right 4, Up 3
C. Right 4, Down 1
D. Right 4, Up 1

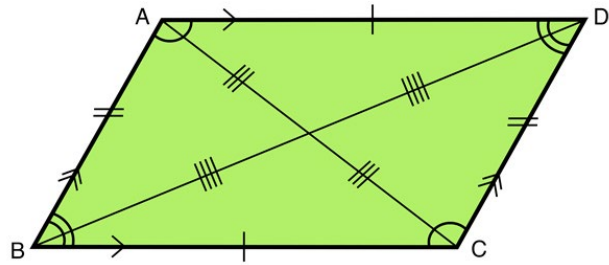
24. Old McDonald had a total of 7 chickens and cows in his pasture. He counted 24 legs in the pasture. How many chickens and how many cows were in the pasture?
- 12 chickens, 0 cows
 - 3 cows, 4 chickens
 - 4 chickens, and 4 cows
 - 5 cows, 2 chickens
25. Which of the following numbers is divisible by both 16 and 3?
- 384
 - 128
 - 450
 - 330
26. What value of x will make this algebraic equation **true**: $\frac{34}{x} = \frac{17}{2}$?
- 2
 - 3
 - 4
 - 17
27. Four ballplayers shake hands with every other ballplayer once. How many total handshakes are there?
- 5
 - 6
 - 7
 - 8
28. It takes 3 hours and 42 minutes to complete painting a room. If the painter begins painting at 8:25 AM and doesn't take any breaks, what time will he be finished painting?
- 11:67 AM
 - 12:07 PM
 - 11:67 PM
 - 12:07 AM
29. Match the mathematical symbols with their definitions.

x. >	1. Pi (3.14)
y. <	2. Less than
z. π	3. Greater than

- $x - 3, y - 2, z - 1$
- $x - 2, y - 1, z - 3$
- $x - 1, y - 2, z - 3$
- $x - 1, y - 3, z - 2$

30. Identify the type of polygon shown here:

- A. Parallelogram
- B. Rhombus
- C. Rectangle
- D. Square



The next six questions are tiebreaker questions.

31. What type of angles can be formed by the intersection of two lines?

- A. Three acute angles and one obtuse angle
- B. Four acute angles
- C. Two right angles and two obtuse angles
- D. Two right angles and two 90° angles

32. A carpenter cut a 4 foot board into 6 pieces. How long was each cut piece?

- A. $\frac{2}{3}$ foot
- B. 8 feet
- C. $\frac{3}{2}$ feet
- D. 6 inches

33. What is the value of 6 cubed?

- A. 18
- B. 216
- C. 21.6
- D. 36

34. A right triangle has one acute angle that measures 60 degrees. What is the measure of the other acute angle?

- A. 20°
- B. 60°
- C. 30°
- D. 45°

35. You roll two fair dice. What is the probability that the sum shown is 12?

- A. $\frac{1}{36}$
- B. $\frac{1}{6}$
- C. $\frac{1}{12}$
- D. $\frac{1}{4}$

36. How many diagonals can be drawn from one vertex of a regular heptagon?

- A. 14
- B. 12
- C. 7
- D. 4

Mathematics Written Assessment

2023 Elementary District Governor's Cup

1. D
2. A
3. B
4. A
5. C
6. D
7. C
8. D
9. A
10. B
11. C
12. B
13. D
14. C
15. B
16. A
17. B
18. B
19. D
20. D

21. C
22. B
23. D
24. D
25. A
26. C
27. B
28. B
29. A
30. A

TIEBREAKERS

31. D
32. A
33. B
34. C
35. A
36. D